NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

# Week 5 – Demand of Maritime Transport





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10 September 2026

# Learning Objectives

1. Explain the shipping market model which depicts the relationship between demand, supply and freight market.
2. Discuss the factors causing the world or business economy to be cyclical in nature.
3. Discuss how random shocks affects seaborne commodity trades.

# Learning Objectives

4. Discuss how seaborne commodity trades is affected by short-term volatility and long-term trends.
5. Calculate the ton-miles and average haul of a commodity.
6. Discuss how transport costs influence purchasing decisions for customers.




# 1. Shipping Market Model

# 1. Shipping Market Model

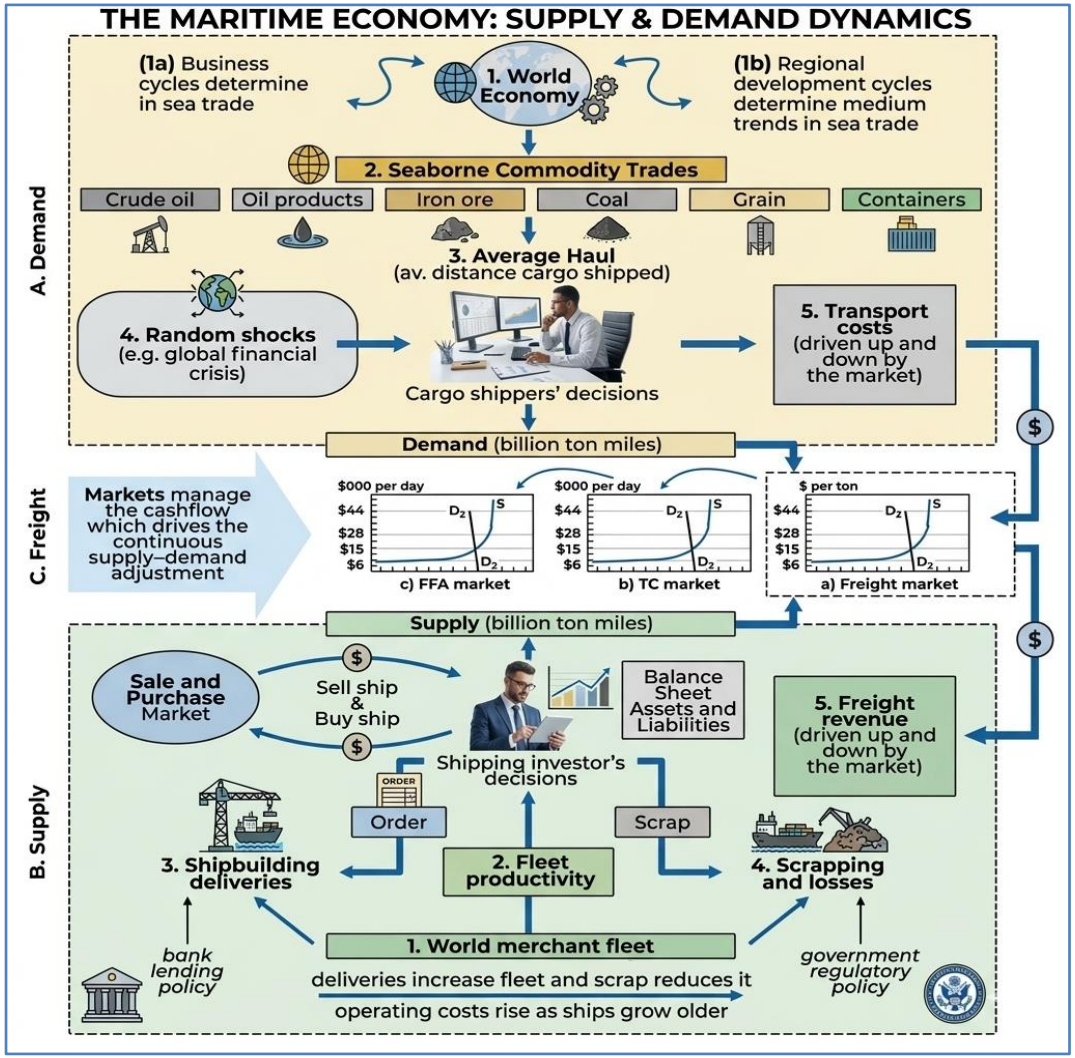
- Aim of shipping companies is to take advantages of the cycles to buy ships at low prices and sell at high prices.
- Only can be achieved if shipping companies are able to recognize or predict the peaks and troughs on the freight market.
- This and next topic explores the mechanisms which determine freight rates > result of demand and supply balance.

# 1. Shipping Market Model

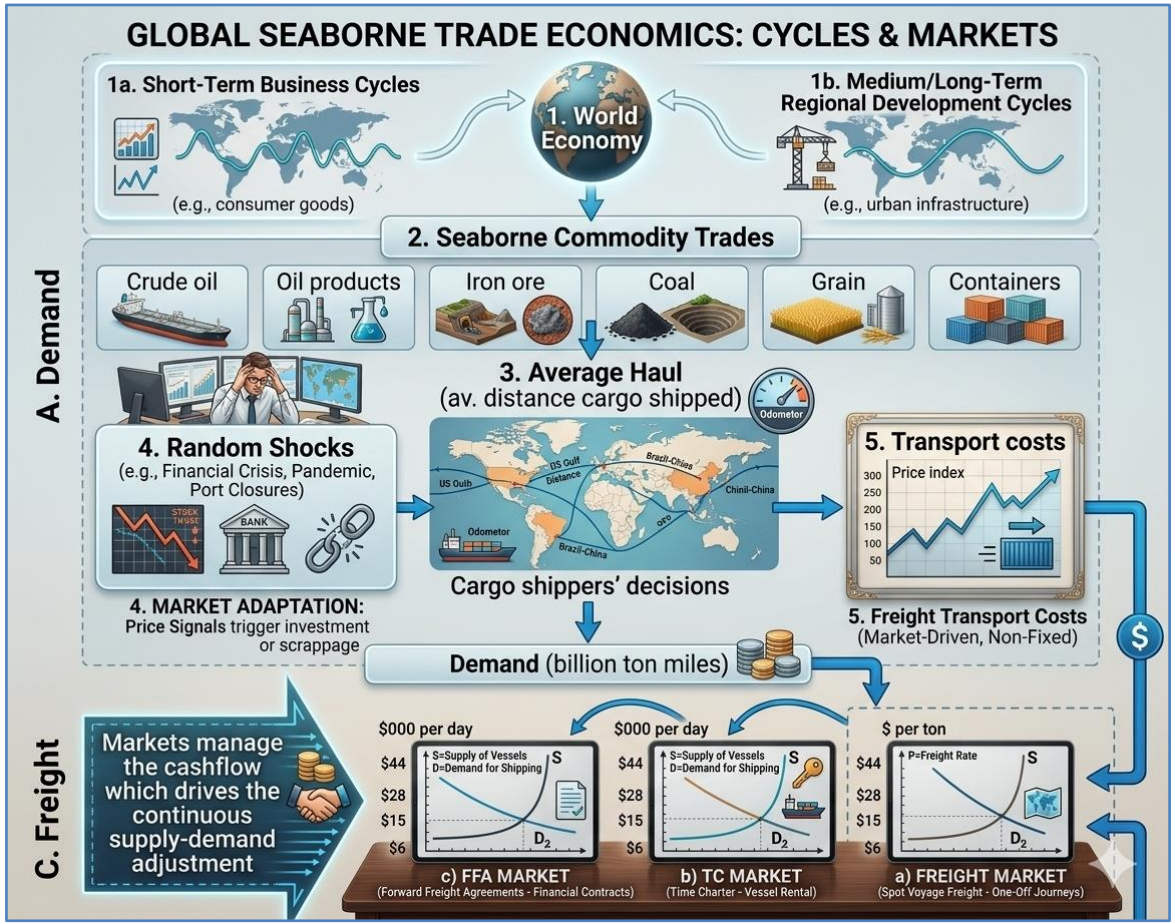
- Five factors affecting the demand for sea transport and five affecting supply.
- These factors fit together to form a model with freight market regulating cash flow from one sector to another.

| Demand | Supply |
|---|---|
| 1. The world economy | 1. World fleet |
| 2. Seaborne commodity trades | 2. Fleet productivity |
| 3. Average haul | 3. Shipbuilding production |
| 4. Random shocks | 4. Scrapping and losses |
| 5. Transport costs | 5. Freight revenue |

# 1. Shipping Market Model



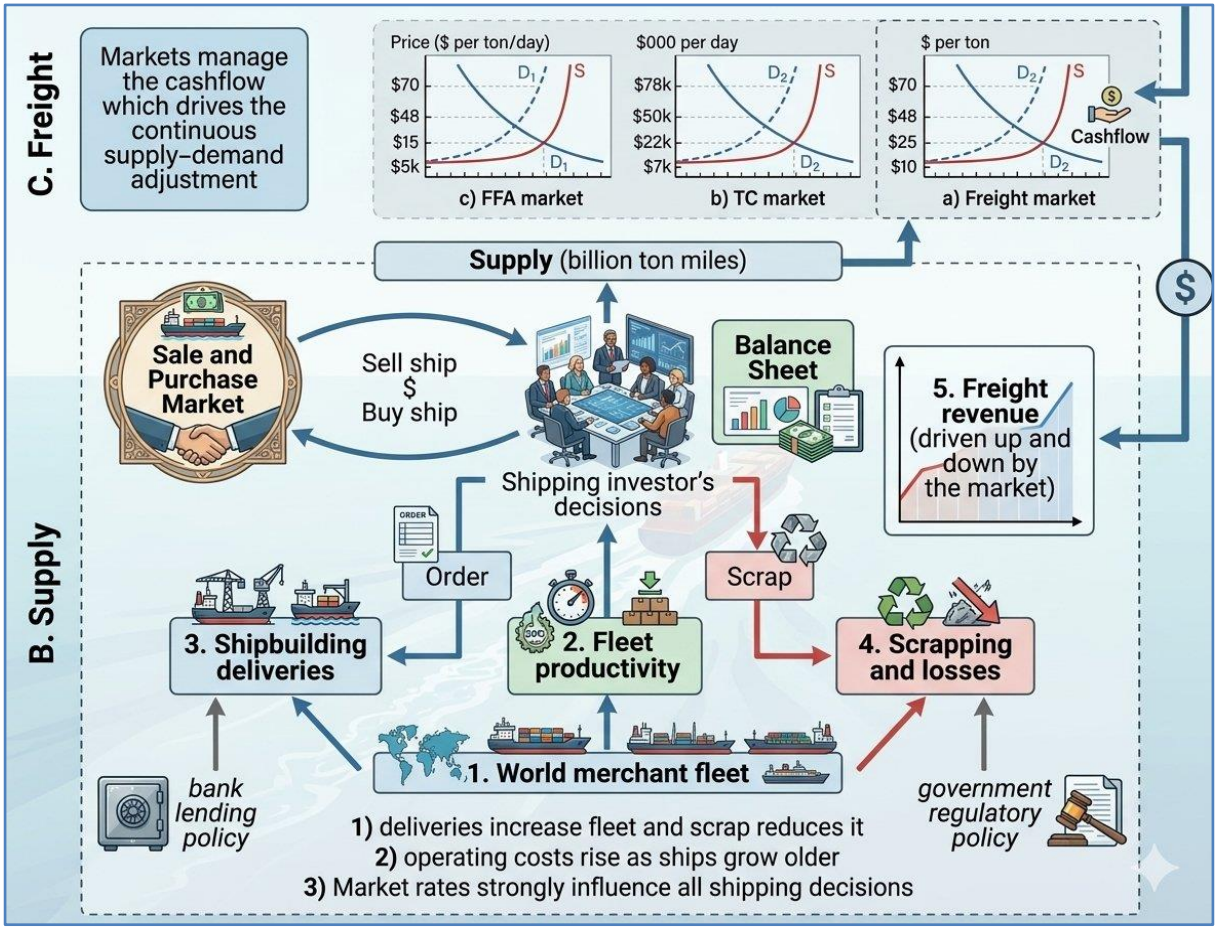
# 1.1 Demand Module (A)



# 1.1 Demand Module (A)

- The _____, through business cycles and regional growth trends, determines the volume of goods traded by sea.
- Developments in particular _____ may modify growth trends (E.g., *development of construction industry in China – influence on steel*)
- Random shocks will influence ton-miles and average haul, and transport costs.

# 1.2 Supply Module (B)



# 1.2 Supply Module (B)

- World merchant fleet provides a fixed stock of transport capacity.
- When demand is low, ships may be laid up.
- Amount of transport depends on ship's efficiency – speed and waiting time.
- Policies of banks and regulators also have an impact (E.g., *amount of lending and environment policy*)

# 1.3 Dynamic Links

- Heart of this shipping market model is *people* and *psychology*.

| People | Examples |
|---|---|
| Cargo Shippers (Demand side) | - Large corporations trading in raw materials (E.g. Cargill) |
| Shipping Investors (Supply side) | - KG companies who owns containerships.<br>- Major oil companies with their own fleets. |

# 1.3 Dynamic Links

- Shipping investors and cargo shippers have to negotiate the rates for each ship.
- Psychology plays a role at the negotiating table
  - rates agreed vary depending on how the negotiating parties feel.
  E.g., *a ship might be fixed for $20,000 per day on Monday but a sister ship might be fixed for $30,000 per day on Tuesday because the charterer might have heard a rumour about charter rate rising*.

# Review Questions

1. Which of the following is **NOT** part of the factors in the demand module in the shipping freight model?

a. Average haul

b. Random shocks

c. Transport costs

d. Freight revenue

**Answer:**

LO2

# 2. World Economy (Business Cycle and Trade Development Cycle)

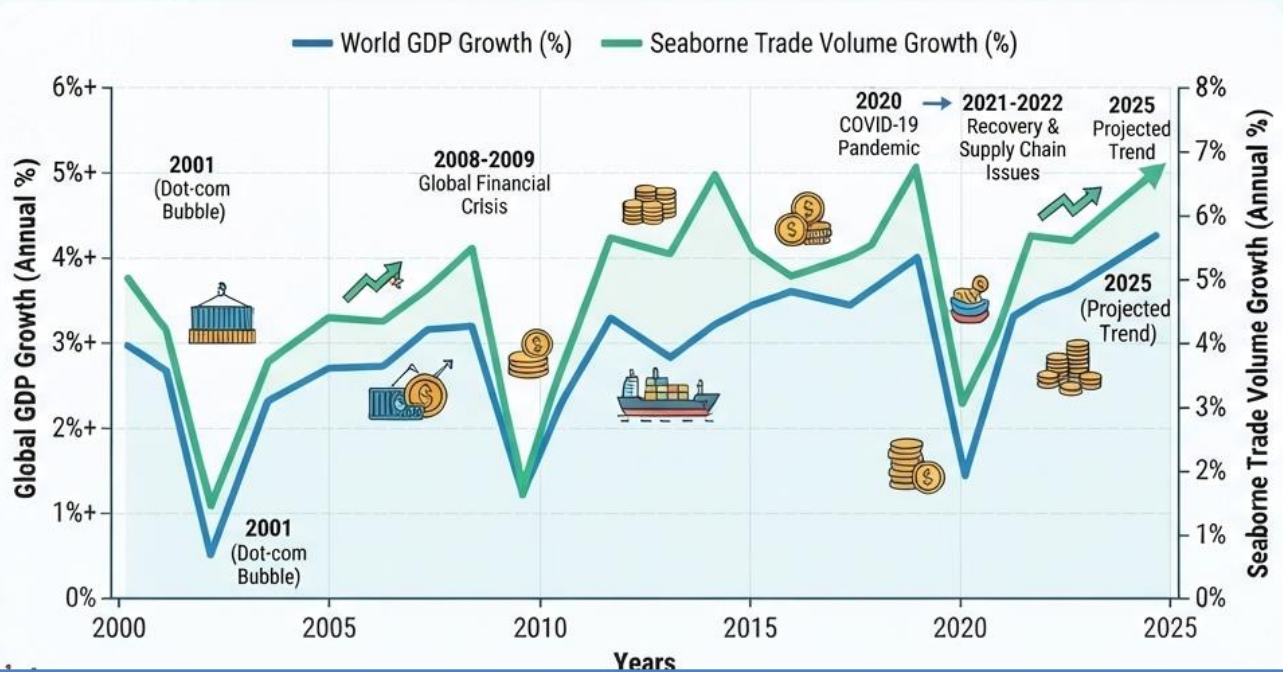
# 2. World Economy

- World economy generates most of the demand for sea transport, through either import of raw materials for manufacturing industry or trade of manufacture products.
- Business cycle determines the short-term cycles in seaborne trade.
- Fluctuations in rate of economic growth influence seaborne trade > creating cyclical pattern of demand.

# World GDP and Seaborne Trade (2000-2020)

## WORLD GDP & SEABORNE TRADE GROWTH: 2000-2025 (Relationship & Trends)

### Key Trend Analysis



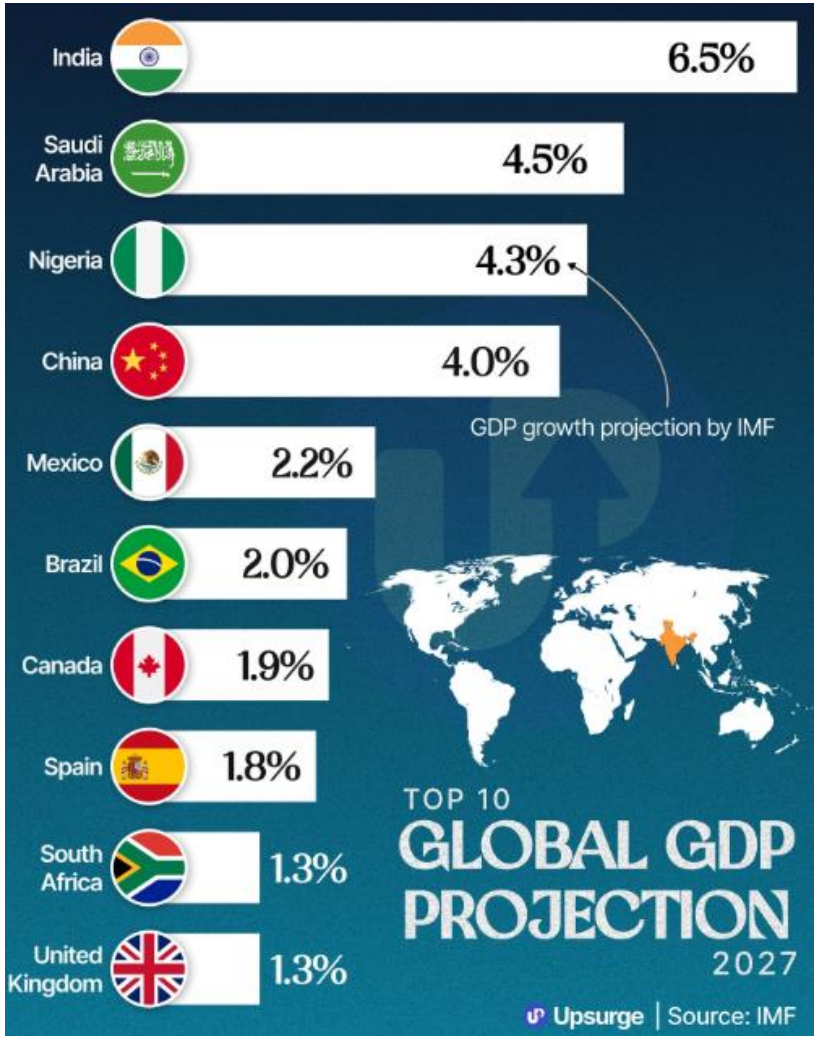

### OBSERVATIONS & CORRELATION

- **Positive correlation:** Global economic expansion generally drives increased seaborne trade.
- **Higher Volatility:** Seaborne trade volume growth tends to be more volatile than GDP growth.
- **Economic Downturns:** Both metrics experience significant declines during global crises (e.g., 2009, 2020).
- **Trade as a Leading/Coincident Indicator:** Reflects shifts in production and consumption.
- **Globalisation:** Deep integration means world trade grows alongside GDP.

# Real GDP growth projections

%, year-on-year

| | 2026 | 2027 |
|---|---|---|
| India | 6.1% | 6.4% |
| Indonesia | 4.8% | 5.0% |
| China | 4.4% | 4.3% |
| Saudi Arabia | 4.0% | 3.6% |
| Türkiye | 3.3% | 3.8% |
| G20 | 3.0% | 3.0% |
| Argentina | 2.8% | 3.5% |
| Australia | 2.3% | 2.4% |
| United States | 2.0% | 1.7% |
| Korea | 1.7% | 2.1% |
| Brazil | 1.5% | 2.1% |
| Mexico | 1.3% | 1.7% |
| Canada | 1.2% | 1.7% |
| South Africa | 1.2% | 1.7% |
| Japan | 0.9% | 0.9% |
| France | 0.8% | 1.0% |
| Germany | 0.8% | 1.5% |
| United Kingdom | 0.7% | 1.3% |
| Russia | 0.6% | 0.8% |
| Italy | 0.4% | 0.6% |



# 2. World Economy

- The relationship between world economy and shipping demand is not simple or direct because of the following aspects:

  a) **Business cycle**

  b) **Trade development cycle**

# 2.1 Business Cycle

- 4 most commonly quoted causes of business cycle:
    1. **Multiplier and accelerator**
    2. **Time-lags**
    3. **Stock-building**
    4. **Mass psychology**

# 2.1.1 Multiplier and Accelerator

- An increase in investment (e.g., *new factories set up*) creates new consumer demand from the workers hired.
- Workers spend their wages > creates more demand (_____ multiplier).
- As extra consumer expenditure trickles through the economy, growth picks up, generating demand for even more investment.

# 2.1.1 Multiplier and Accelerator

- Eventually, capital becomes fully utilized and economy over-heats.
- Expansion sharply halted, throwing whole process into reverse.
- Investment falls off, jobs are lost, and multiplier and accelerator go into reverse > recession.

# 2.1.2 Time Lags

- Delays between economic decisions and their implementation make cyclical fluctuations more extreme.
- _____ – excessive interest rates cuts to fix slow economy .
- _____ – businesses make plans based on current conditions but regulations/money supplies changing today will only hit their profits much later.

# 2.1.3 Stock-building

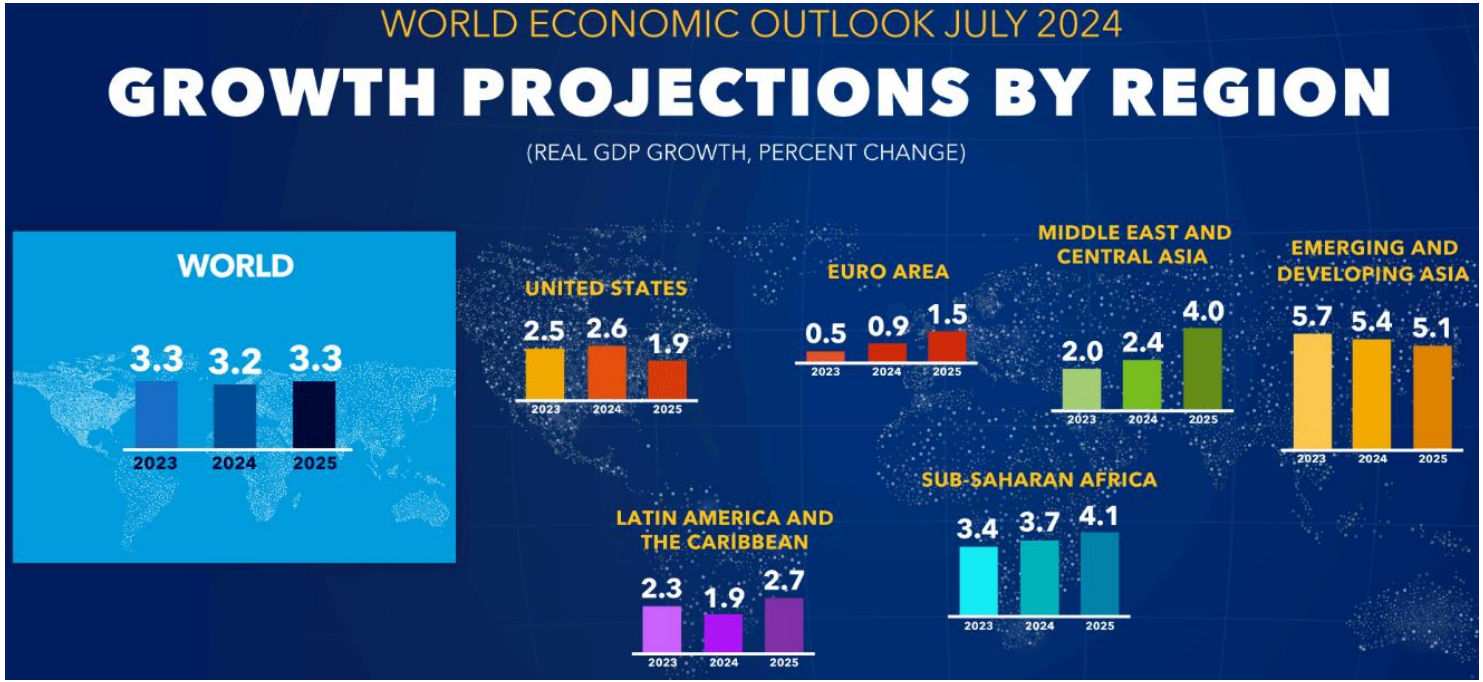
- Stock-building produces sudden bursts of demand as industries adjust their stocks/inventory during the business cycle.
- During recessions, financially hard-pressed manufacturers stop ordering new inventory, intensifying downturn in demand for sea transport.
- When economy recovers, sudden rush to rebuild inventory > sudden burst in commodity demand.

# 2.1.4 Mass Psychology

- If they act in an imitative manner, a particular trend will build up where they can affect whole economic system.
- Periods of optimism or pessimism become self-fulfilling through medium of stock exchanges, financial booms and behaviour of investors.

# 2.2 Trade Development Cycle

- Regional development cycles determine the medium to long-term trends in seaborne trade.



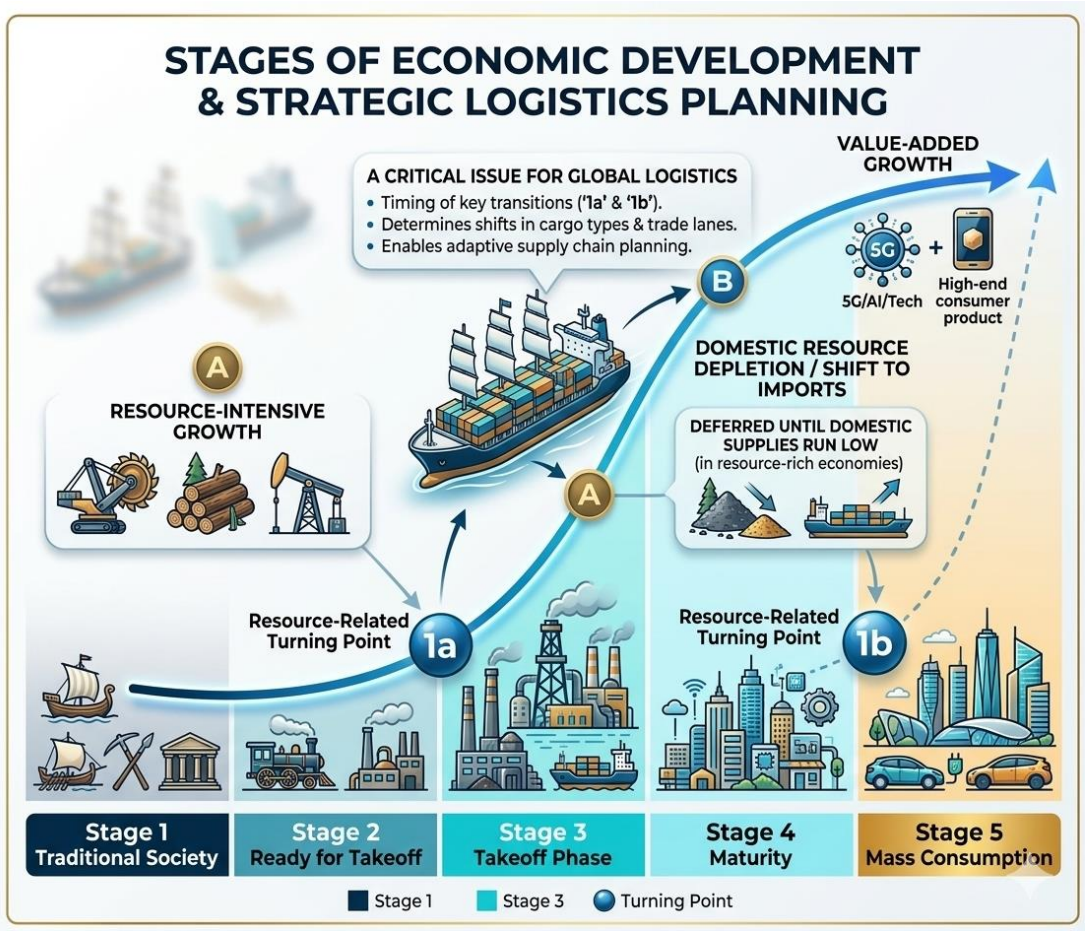
# 2.2 Trade Development Cycle

- Two reasons why, over long periods, trade growth of individual regions will probably change.

  a. Economic structure changes over time.

  b. Ability of local resources of food and raw materials to meet local demand.

# 2.2 Trade Development Cycle



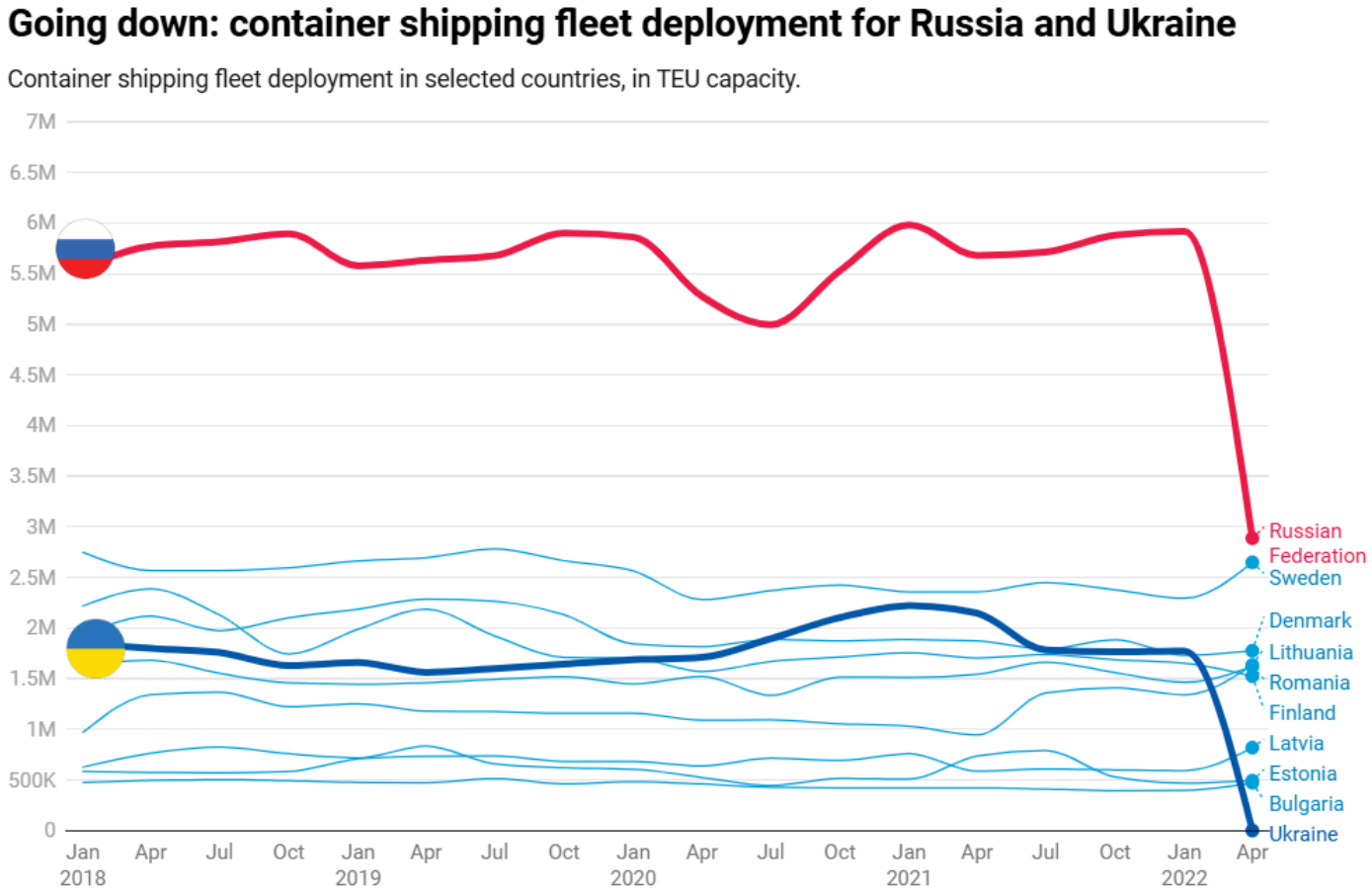



# 3. Random Shocks

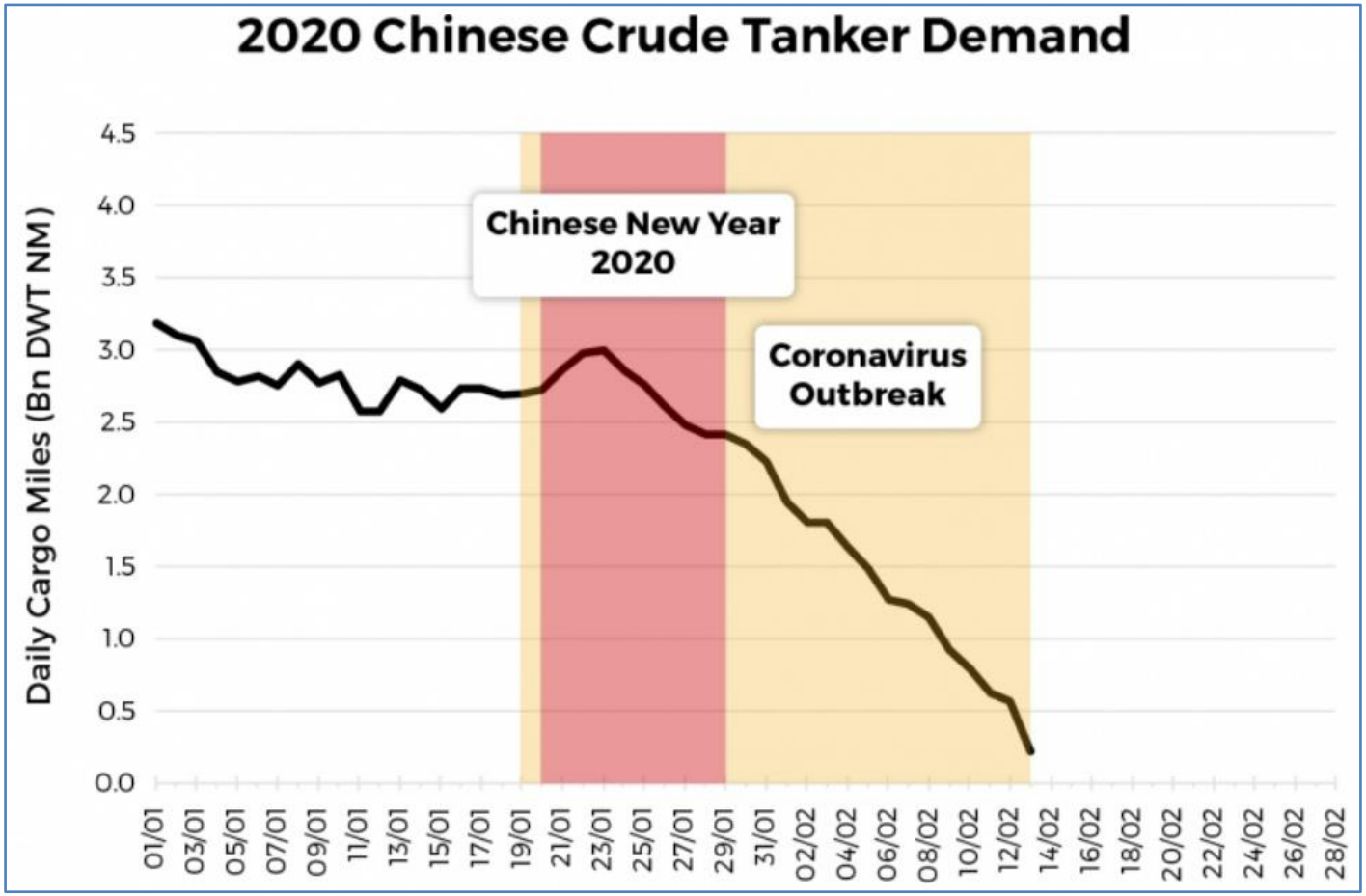
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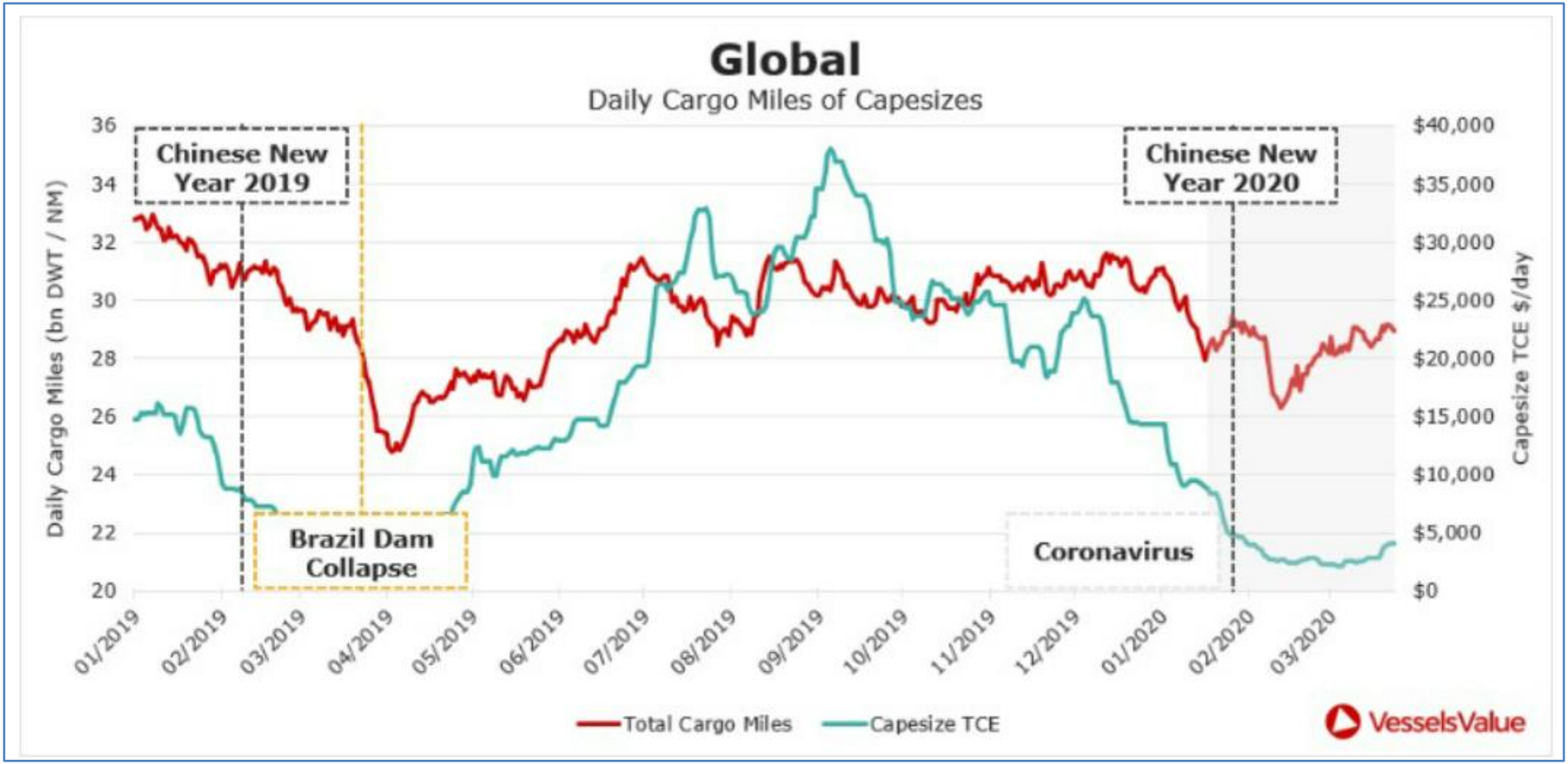
- Upsetting the stability of the economic system due to unexpected shocks.
- Could be political (Russia-Ukraine war), economic (trade war between US and China), new resources (AI), weather changes, pandemics (covid).
- Differs from cycles because they are unique, often precipitated by some particular event and impact on shipping market is often very severe.

# 3.1 Russia-Ukraine War



# 3.2 Covid Pandemic




Global
Daily Cargo Miles of Capesizes
Daily Cargo Miles (bn DWT / NM)
Capesize TCE $/day
Chinese New Year 2019
Brazil Dam Collapse
Chinese New Year 2020
Coronavirus
36
34
32
30
28
26
24
22
20
$40,000
$35,000
$30,000
$25,000
$20,000
$15,000
$10,000
$5,000
$0
01/2019
02/2019
03/2019
04/2019
05/2019
06/2019
07/2019
08/2019
09/2019
10/2019
11/2019
12/2019
01/2020
02/2020
03/2020
Total Cargo Miles
Capesize TCE
VesselsValue

# The price of shipping is rising again

ClarkSea Index in Thousand USD/day– all shipping markets.



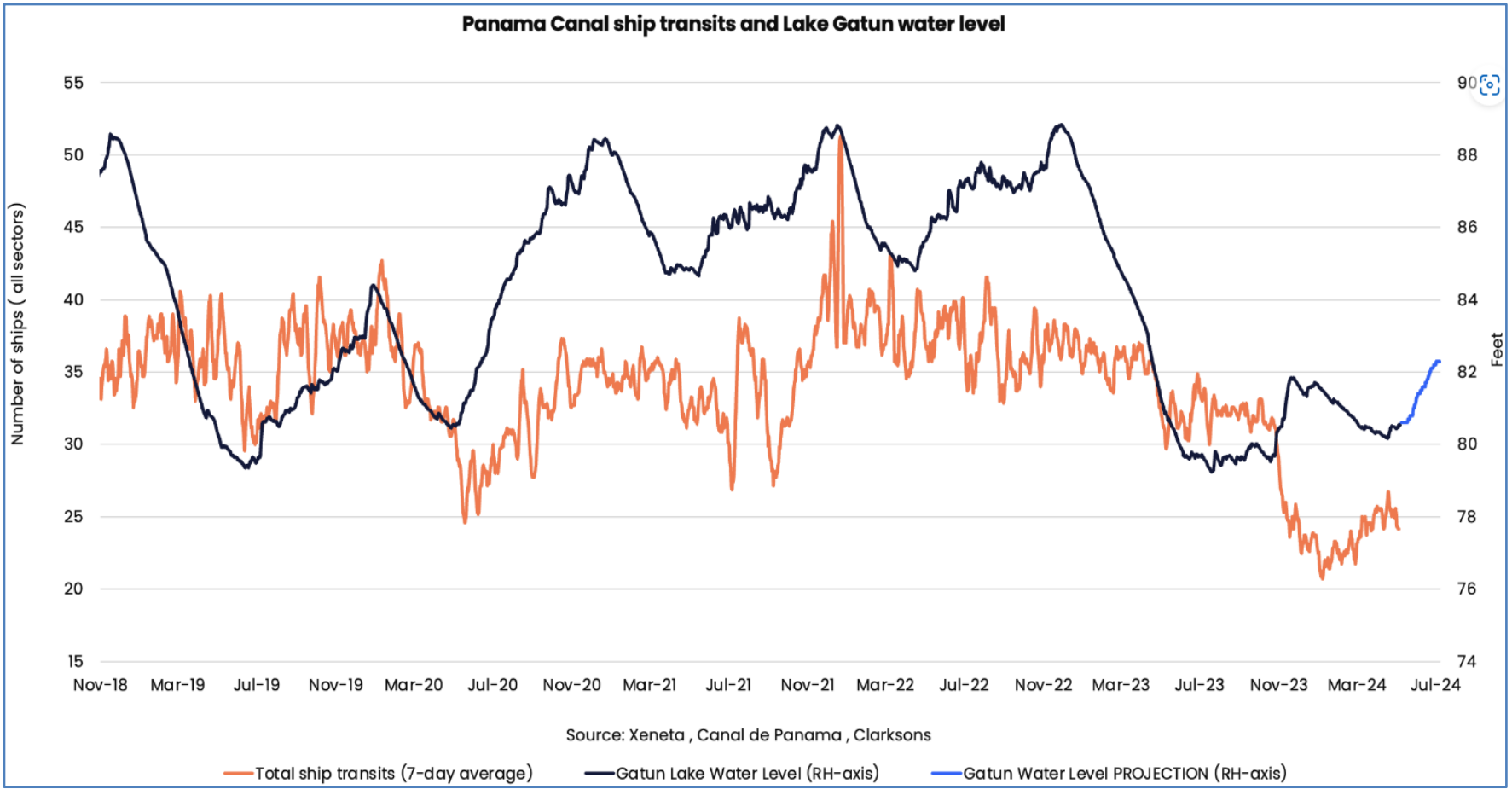
War in Ukraine

COVID-19
pandemic effects

50
45
40
35
30
25
20
15
10
5
0

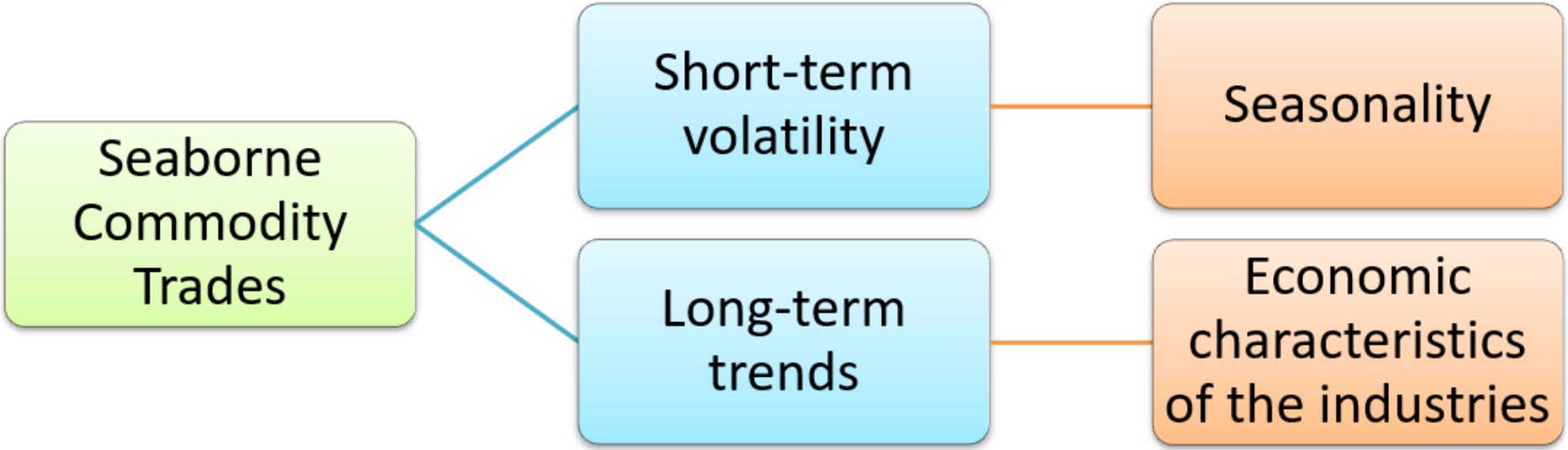
Jan 2019
Apr
Jul
Oct
Jan 2020
Apr
Jul
Oct
Jan 2021
Apr
Jul
Oct
Jan 2022
Apr

# 3.3 Weather Changes



# 4. Seaborne Commodity Trades (Short-Term Volatility and Long-Term Trends)

# 4. Seaborne Commodity Trades



# 4.1 Short-Term Volatility

- An important cause is the _____.
  - many agricultural commodities are subject to seasonal variations caused by harvests.

E.g. *grain, sugar and citrus fruits*.

- Grain exports from the U.S. Gulf reach a trough in the summer then build up in September as the crop is harvested. Trade may increase by as much as 50% between September and end of the year.
- In liner trade, seasonal peaks occur during August to October leading to major holidays.

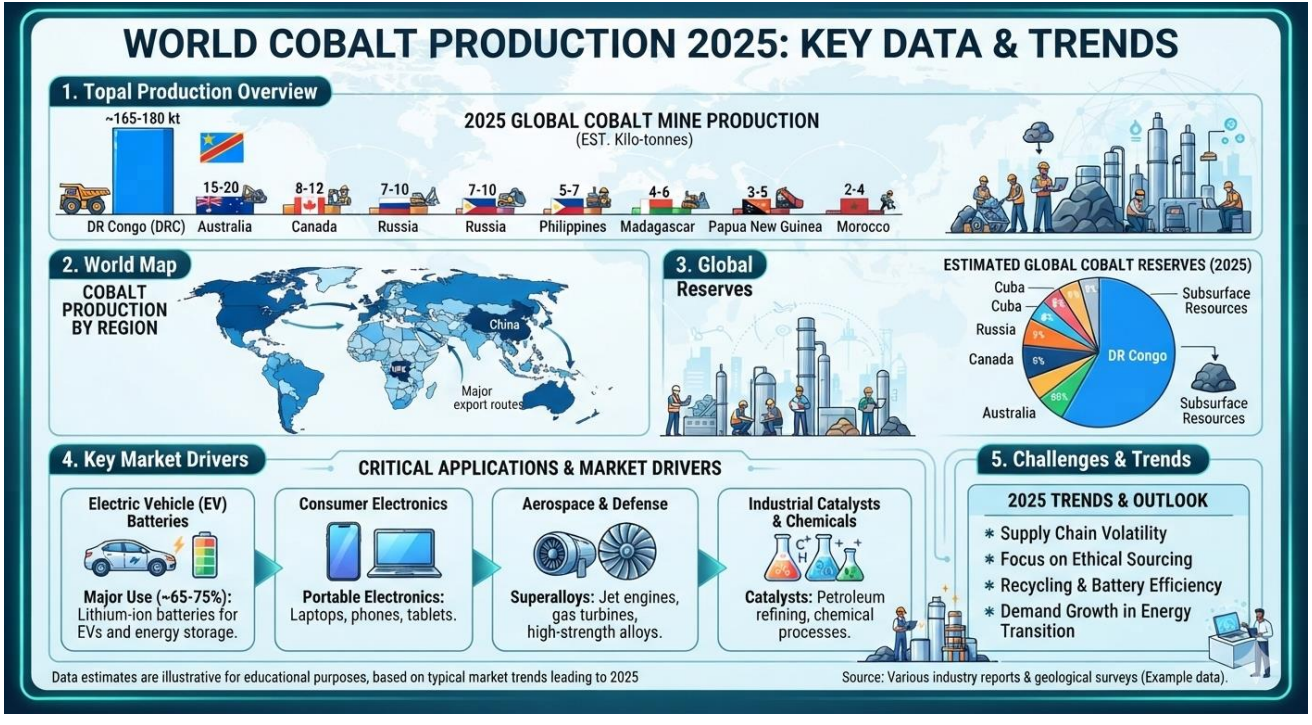
# 4.2 Long-Term Trends

Long-term trends in commodity trade can be identified by observing economic characteristics of the industries that produce and consume the commodities in terms of:

| form | place | process | time |
| --- | --- | --- | --- |
| • i.e., change in demand for particular products; | • i.e., change in sources from which supplies of the commodity are obtained; | • i.e., change due to the relocation of processing plants that change the trade pattern; and | • i.e., change in shippers' requirements to obtain what they desire at designated times. |

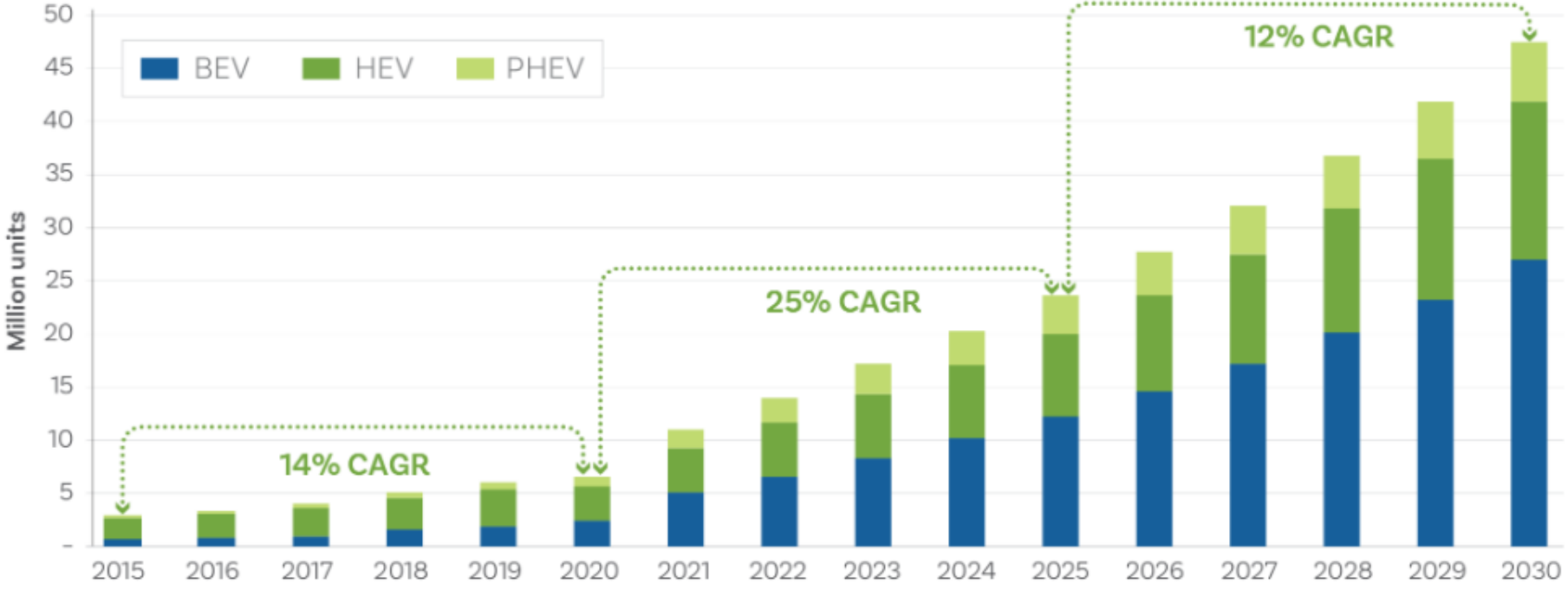
# 4.2.1 Form

- Refers to *changes in demand for a particular commodity*. E.g., Cobalt



Electric vehicles include Battery Electric Vehicles (BEVs) and Plug in Hybrid Vehicles (PHEVs), Hybrid Electric Vehicles (HEVs), commercial trucks, buses, and electric bikes.

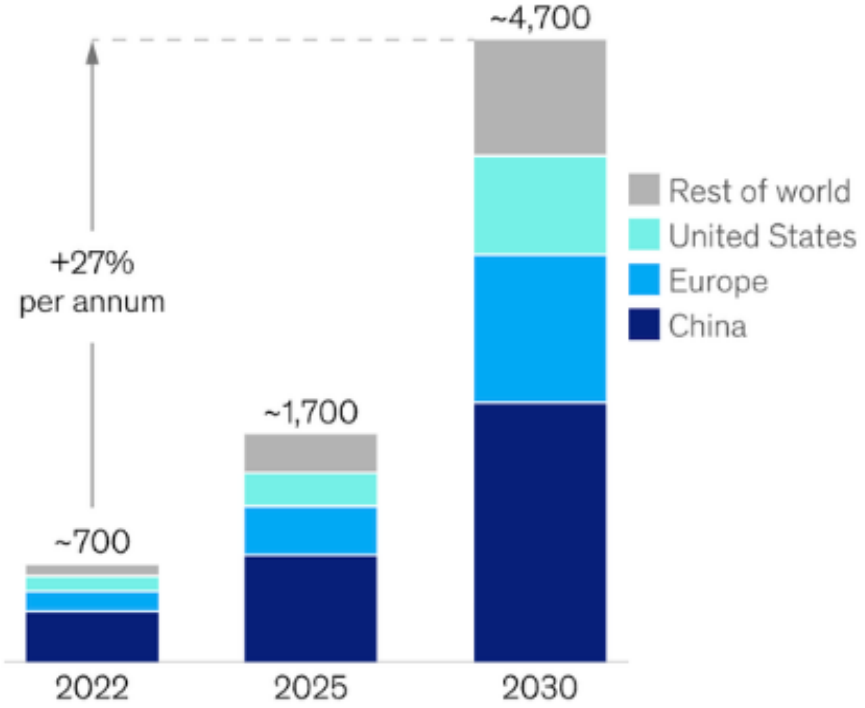
Figure 9 – **EV Sales 2015–2030e**



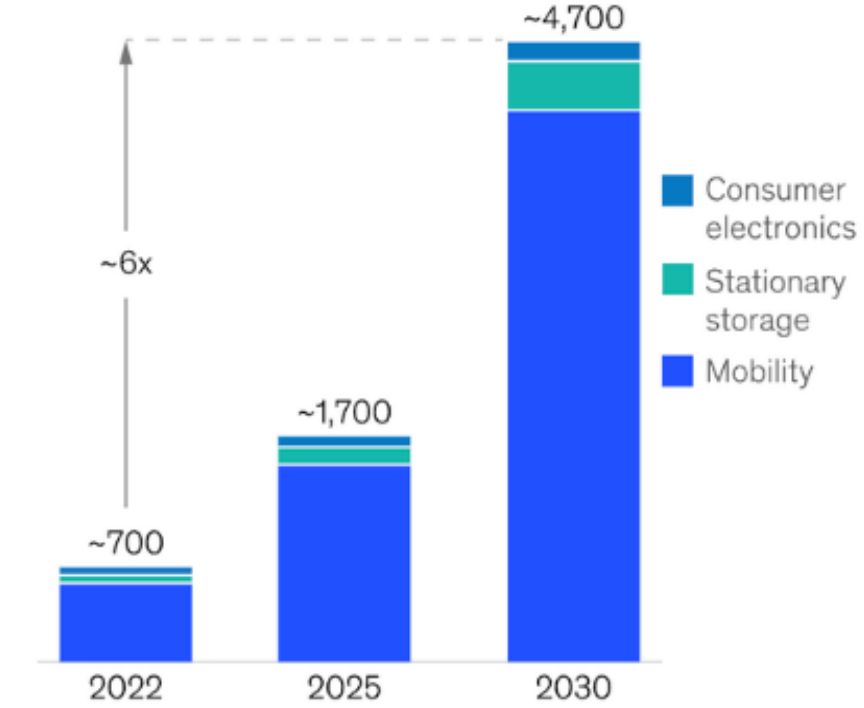

*Source: Wood Mackenzie*

## **Global Li-ion battery cell demand,** GWh, Base case

### By region

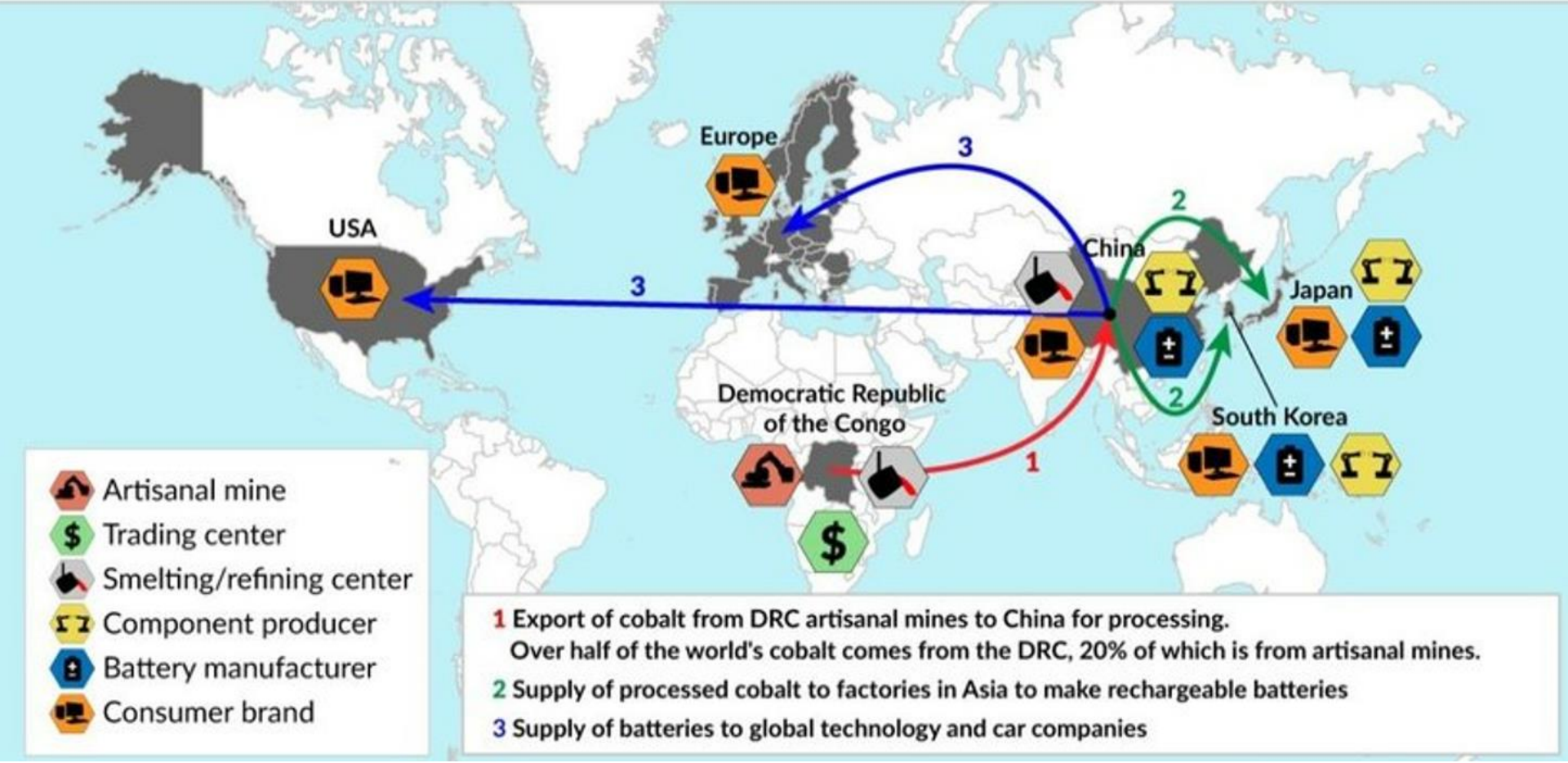



### By sector

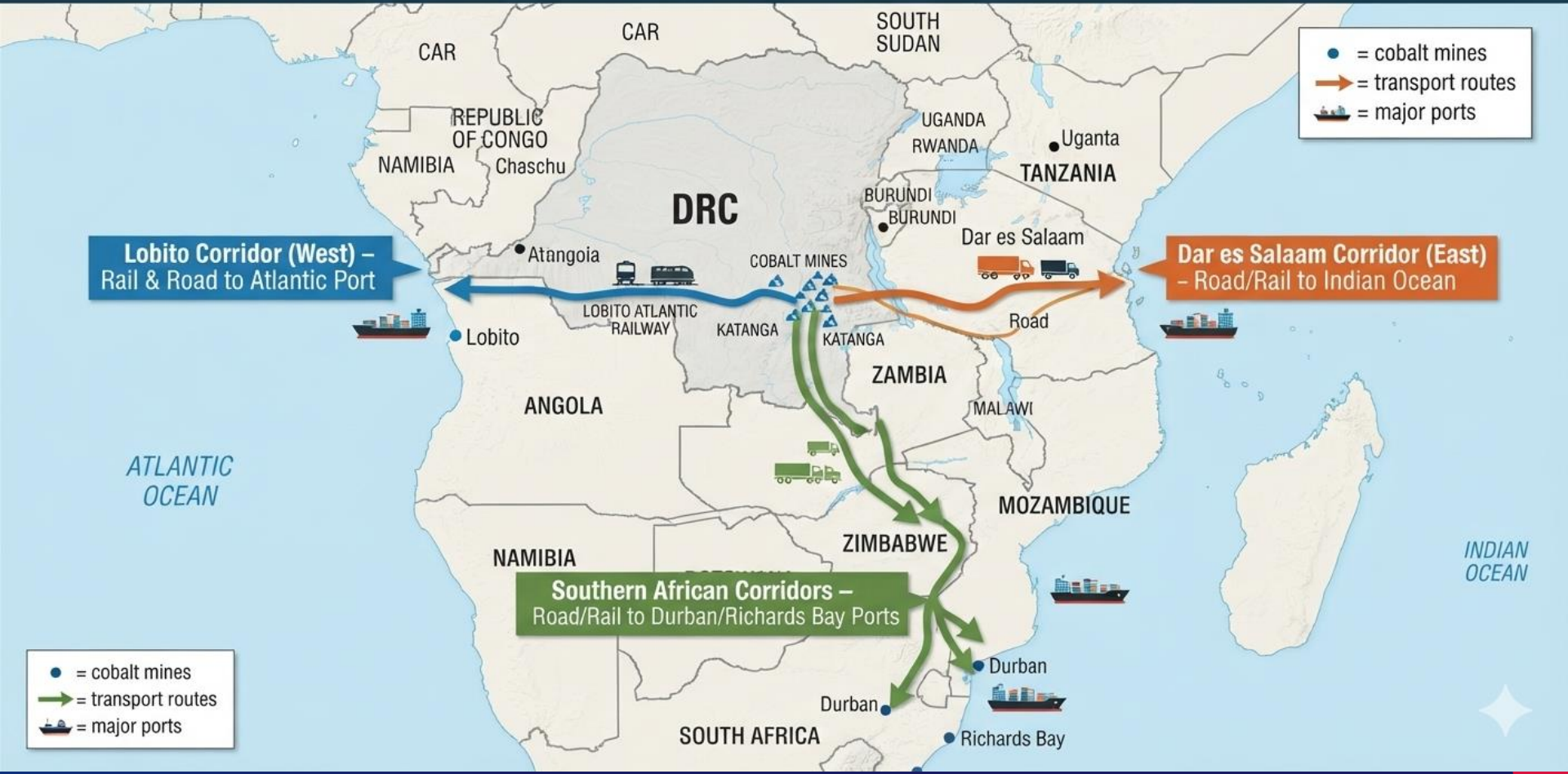



[1]Including passenger cars, commercial vehicles, two-to-three wheelers, off-highway vehicles, and aviation.
Source: McKinsey Battery Insights Demand Model

# Movement of cobalt from artisanal mines in the DRC to the global market



# GLOBAL COBALT SUPPLY CHAIN: TRANSPORT ROUTES FROM THE DRC



Most electric cars are powered by lithium-ion batteries, a type of battery that is recharged when lithium ions flow from a positively charged electrode, called a cathode, to a negatively electrode, called an anode. In most lithium-ion batteries, the cathode contains cobalt, a metal that offers high stability and energy density.

However, cobalt has significant downsides. A scarce metal, its price can fluctuate dramatically, and much of the world's cobalt deposits are located in politically unstable countries. Cobalt extraction creates hazardous working conditions and generates toxic waste that contaminates land, air, and water surrounding the mines.

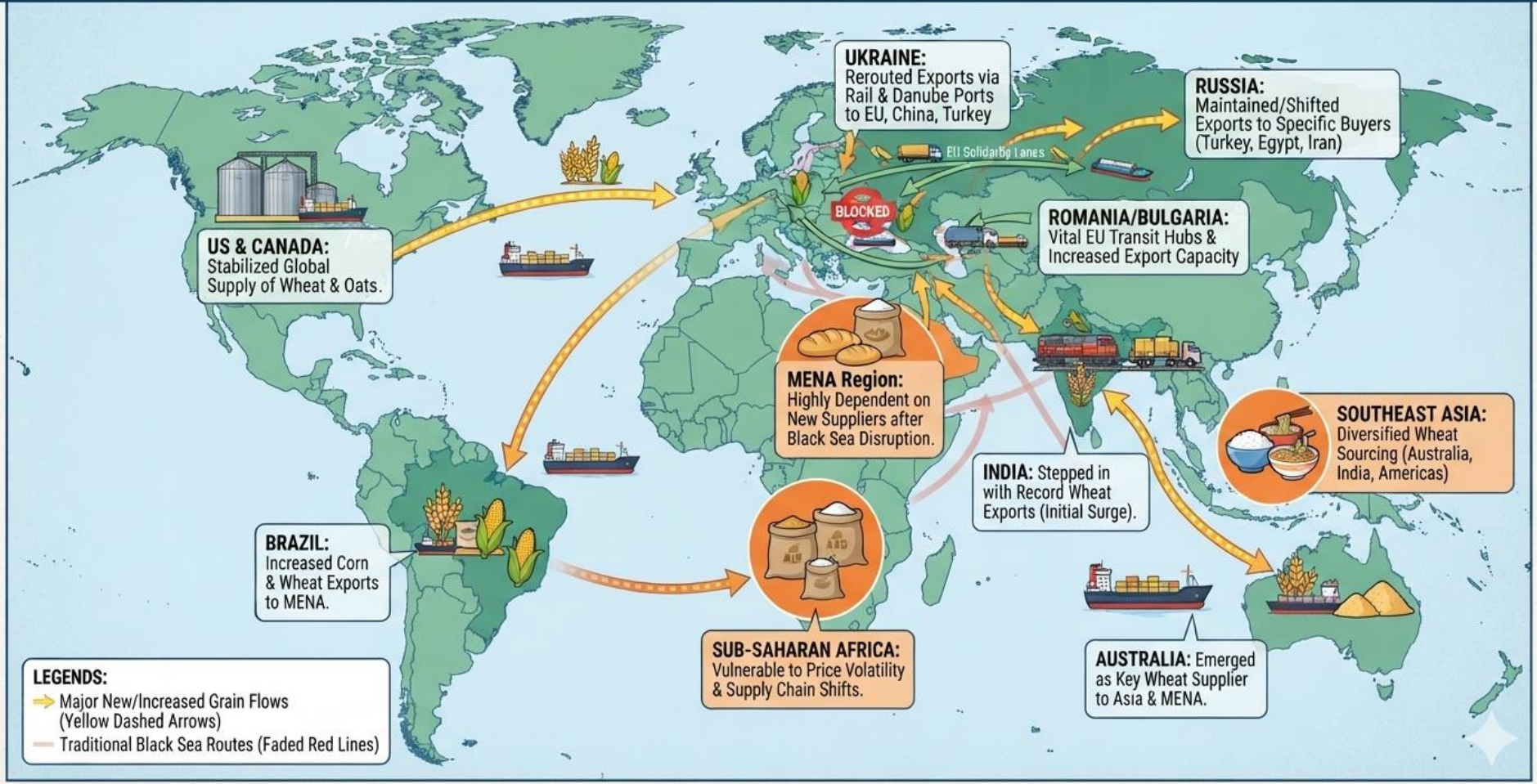
"Cobalt batteries can store a lot of energy, and they have all of features that people care about in terms of performance, but they have the issue of not being widely available, and the cost fluctuates broadly with commodity prices. And, as you transition to a much higher proportion of electrified vehicles in the consumer market, it's certainly going to get more expensive,"

Cobalt-free batteries could power cars of the future | MIT News | Massachusetts Institute of Technology

# 4.2.2 Place

- Refers to *changes in sources from which supplies of commodities are obtained*.
- Affected by geopolitical tensions, climate changes, national protectionism.
- Example:
  (1) grain trade after the Russia-Ukraine war.
  (2) Chinese iron ore import from Australia and Brazil.
  (3) crude oil trade with new technology.

# GLOBAL GRAIN SHIFTS: ALTERNATIVE SOURCING AFTER THE RUSSIA-UKRAINE WAR



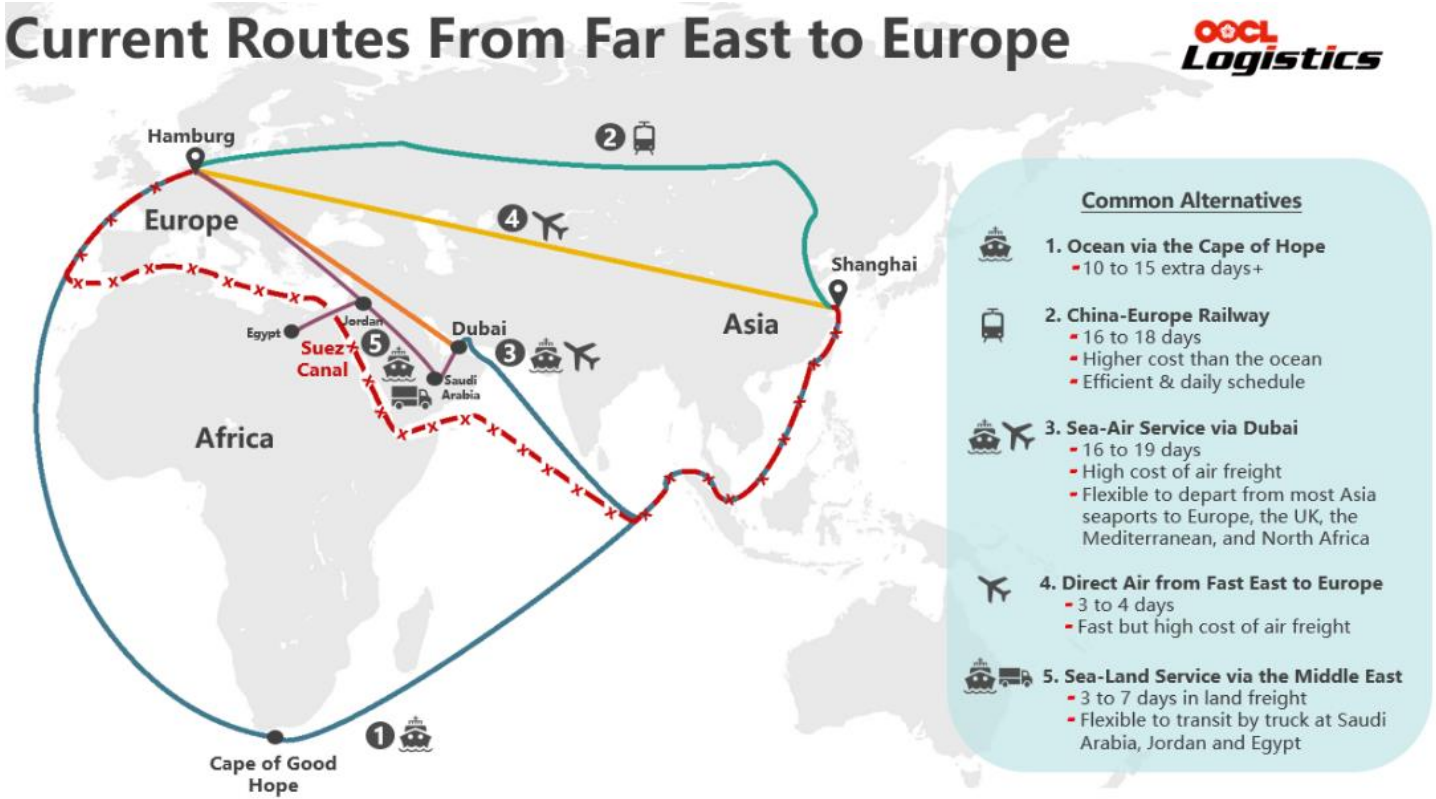
# 4.2.3 Process

- Refers to *changes due to the relocation of processing plants which changes the trade pattern*.
- **Reshoring** – bringing production back to the country or origin or nearby region.
- **Nearshoring** – practice of relocating production to company's primary market or within a nearby region.



# 4.2.4 Time

- Refers *changes in shipper's transport policy*.



# Review Questions

- "*Changes due to the relocation of plants which changes trade patterns*" refers to which of the long-term trends?

a. Form

b. Place

c. Process

d. Time

**Answer:**

# 5. Ton-Miles and Average Haul

# UNDERSTANDING TON-MILES AND AVERAGE HAUL

Key Metrics in Freight Transportation

## TON-MILES

A measure of freight transport output that combines the weight (in tons) and the distance (in miles) cargo is carried.

[ [TONNAGE (WEIGHT)] x DISTANCE = TON-MILES ]



1. Short Haul, Heavy Load:
   100 Tons x 500 Miles
   = 50,000 Ton-Miles
2. Medium Haul:
   50 Tons x 1,500 Miles
   = 75,000 Ton-Miles
3. Long Haul, Light Load:
   20 Tons x 5,000 Miles
   = 100,000 Ton-Miles

**Key Takeaway:** Reflects Transport Volume & Work Done

## AVERAGE HAUL

The average distance a single ton of cargo is transported, calculated by dividing Ton-Miles by Total Tons.

$$\text{AVERAGE HAUL} = \frac{\text{TOTAL TON-MILES}}{\text{TOTAL TONS}}$$

### Calculating Average Haul

Route A (Short):
$\frac{50{,}000 \text{ Ton-Miles}}{100 \text{ Tons}}$ = 500 Miles

### HISTORICAL TRENDS BY SHIPPING SECTOR

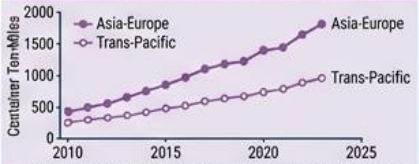
(Real-World Data Insights)

#### DRY BULK SECTOR




- Fluctuates with commodity demand (China's urbanization). Historical highs in late 2010s.

#### CONTAINER SHIPPING SECTOR




- Highest growth rate due to globalized manufacturing and consumer demand. Accelerated by e-commerce.

FREIGHT ECONOMICS

SHIPPING METRICS

# 5. Ton-Miles and Average Haul

- Ton-miles = tonnage of cargo shipped x average distance over which its transported.
- Average haul = ton-miles ÷ tons of freight carried.
  - E.g., in 2024, three shipments (n=3) of crude oil were performed

| Shipment | Weight (mt) | Distance (miles) | Ton-miles |
|---|---|---|---|
| A | 5 | 20 | 100 |
| B | 10 | 80 | 800 |
| C | 7 | 100 | 700 |

  - Average haul = $(\sum ton - miles) \div \sum \text{tons}$

$= (100 + 800 + 700) \div (5 + 10 + 7)$

$= 72.73$

# Review Questions

- Based on the data below, what is the average haul for the shipment of crude oil?

| Shipment | Weight (mt) | Distance (miles) | Ton-miles |
|---|---|---|---|
| A | 8 | 20 | |
| B | 12 | 40 | |
| C | 14 | 60 | |

# 5. Ton-Miles and Average Haul

**There are two routes to transport cargo:**

Route A: Weekly demand for the cargo is 100,000 tons. Each ship is able to carry 50,000 tons of cargo. Round-trip time of this route is **35 days**.

Route B: Weekly demand for the cargo is 100,000 tons. Each ship is able to carry 50,000 tons of cargo. Round-trip time of this route is **70 days**.

**How many ships are required for route A and route B?**

- Demand for shipping service:

  =

- Number of ships require for route A:

  =

  =

- Number of ships require for route B:

  =

  =

# 6. Transport Costs

# 6. Transport Costs

**Consumers make purchase decisions based on transport costs and product qualities.**

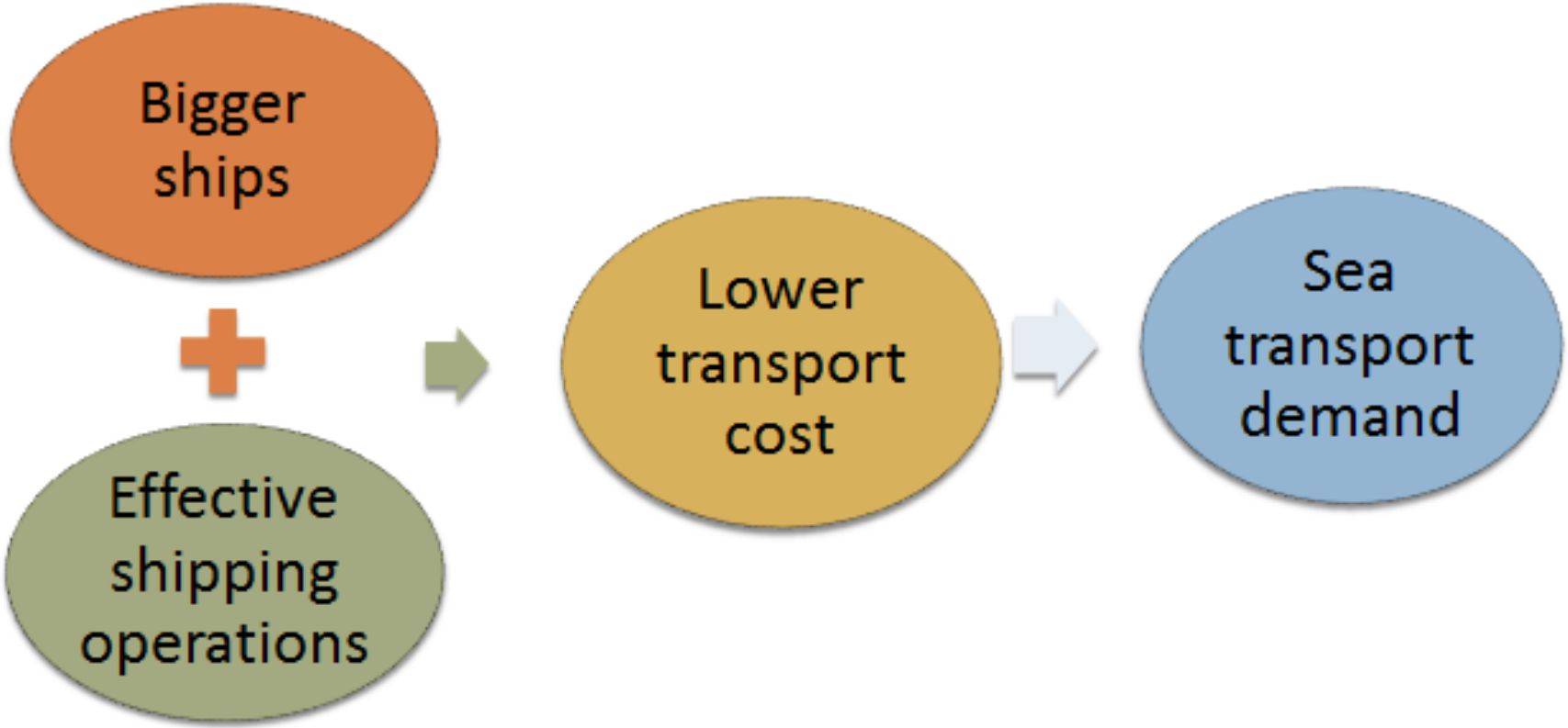
**Product decisions (such as what products to produce or where to distribute) are linked to transport costs and availability of transport services.**

**Decisions on where to market** the products are largely affected by the ability of transport operators to deliver products to markets at a low cost.

**Sourcing decisions** on where to buy raw materials or finished goods consider transport costs.

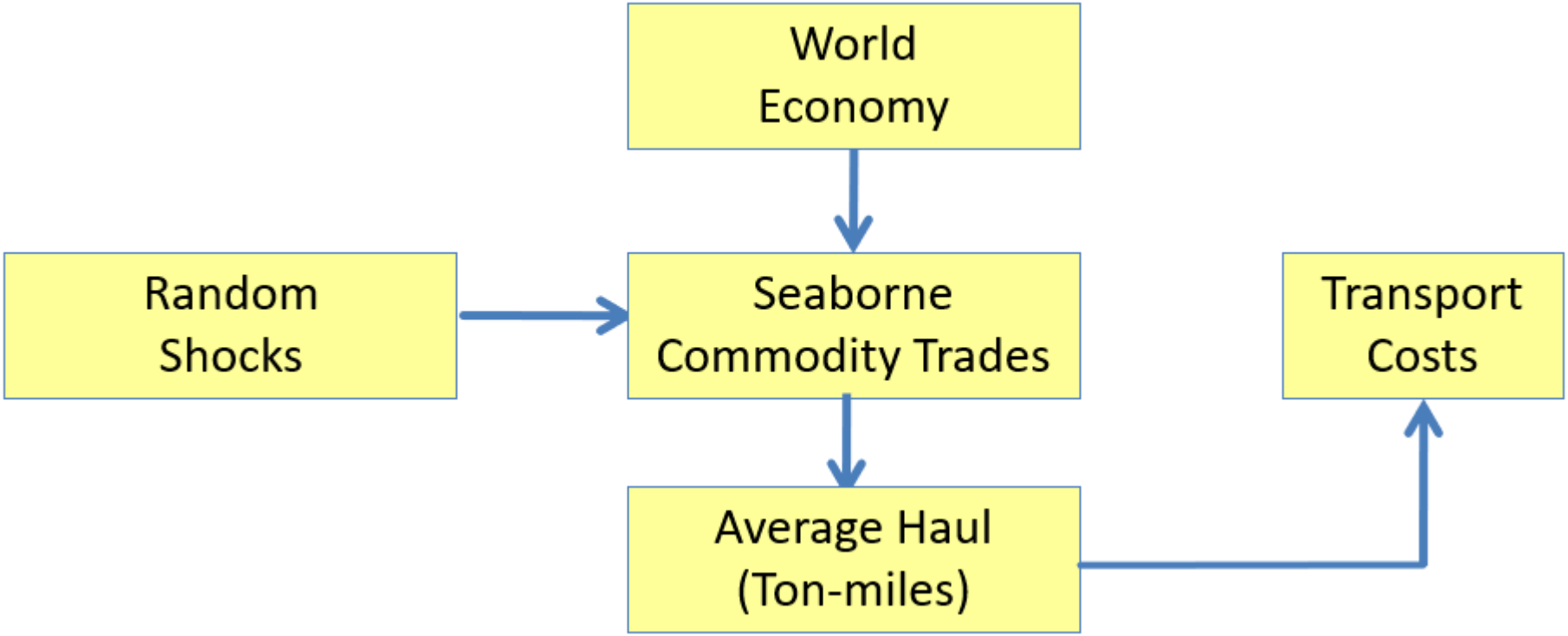
**Pricing decisions** are largely affected by transport costs, which can exert an influence on seaborne trade and long-term trade development.

# 6. Transport Costs



# Summary

# Summary


World Economy
Random Shocks
Seaborne Commodity Trades
Transport Costs
Average Haul (Ton-miles)